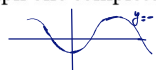


Warm up

1. Graph one complete cycle of $y = -3\cos\left(\frac{1}{4}\left(\theta + \frac{\pi}{3}\right)\right) + 1$.



The base graph of $y = \cos \theta$ is reflected across x-axis & translated

$$a = -3$$

$$|a| = 3$$

$$k = \frac{1}{4}$$

$$d = -\frac{\pi}{3}$$

$$c = 1$$

$$P = \frac{2\pi}{|k|}$$

$$= \frac{2\pi}{1/4}$$

$$P = 8\pi$$

$$\frac{P}{4} = \frac{8\pi}{4}$$

$$\frac{P}{4} = 2\pi$$

$$\text{max} = c + |a|$$

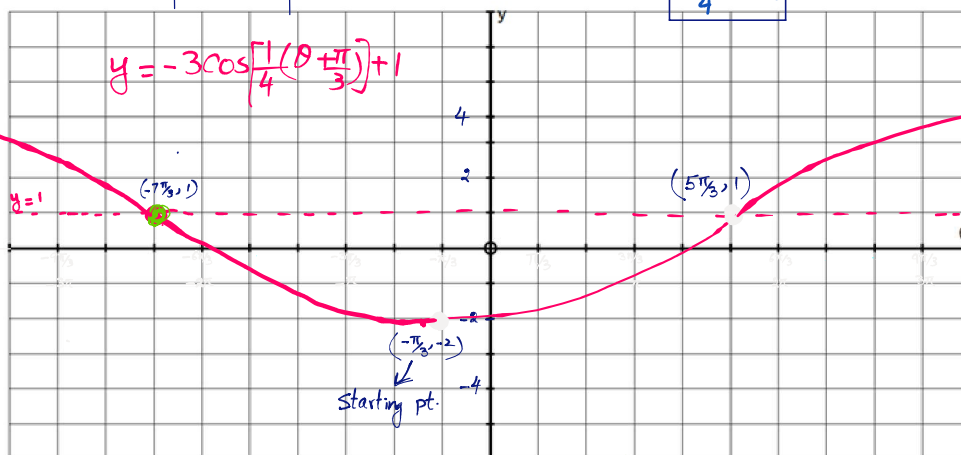
$$= 1 + 3$$

$$\text{max} = 4$$

$$\text{min} = c - |a|$$

$$= 1 - 3$$

$$\text{min} = -2$$



Scale: LCM of d & $\frac{P}{4}$

$$\Rightarrow \frac{\pi}{3}, 2\pi$$

$$\Rightarrow \frac{2\pi}{3}$$

Key pts = starting pt $\pm$ $\frac{\text{Period}}{4}$

$$-\frac{\pi}{3} + 2\pi = \frac{5\pi}{3}$$

$$\frac{5\pi}{3} + 2\pi = \frac{11\pi}{3}$$

$$-\frac{\pi}{3} - 2\pi = -\frac{7\pi}{3}$$

$$-\frac{7\pi}{3} - 2\pi = -\frac{13\pi}{3}$$

2. The graph of $y = 6\cos\left(x + \frac{3\pi}{4}\right) + 1$ is illustrated below. Determine the exact values of g, h, m and n .

$$a = 6$$

$$k = 1$$

$$d = -\frac{3\pi}{4}$$

$$c = 1$$

$$\text{max} = c + |a|$$

$$= 1 + 6$$

$$\text{max} = 7$$

$$\text{min} = c - |a|$$

$$= 1 - 6$$

$$\text{min} = -5$$

new period = $\frac{\text{original period}}{|k|}$

$$= \frac{2\pi}{1}$$

$$= 2\pi$$

m = first min or right

$$= \frac{\text{Period}}{2} - \text{Phase shift}$$

$$= \frac{2\pi}{2} - \frac{3\pi}{4}$$

$$= \frac{4\pi - 3\pi}{4}$$

$$m = \frac{\pi}{4}$$

$$g = -\frac{3\pi}{4}$$

$$h = \text{max}$$

$$h = 7$$

$$n = \text{min}$$

$$n = -5$$

